

INVESTMENTS

Ninth Canadian Edition · Bodie, Kane, Marcus

CHAPTER 1

The Investment Environment

Verified against source text — learning objectives, formulas, and worked examples checked directly against the chapter.

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Study Guide · Prepared for Personal Use

LO1 · Real Assets vs. Financial Assets

The material wealth of a society is determined by the productive capacity of its economy — the **real assets** (land, buildings, machines, knowledge) that produce goods and services. **Financial assets** (stocks, bonds) are claims on the income generated by real assets. They do not directly add to productive capacity; they simply define who owns the real assets and therefore who receives the income they produce.

Financial assets always appear on **both sides** of balance sheets: your claim on a firm is your asset, but the firm's issuance of that claim is its liability. When we aggregate over the whole economy, financial assets **cancel out**, leaving only the sum of real assets as net national wealth. Financial assets are also created and destroyed in the ordinary course of business (e.g., when a loan is repaid); real assets are destroyed only by accident or wear.

Statistics Canada data show that about **47% of Canadian household wealth** consists of non-financial (real) assets. Of that, residential structures are the largest category (~37%), followed by consumer durables (~11%). On the financial side, equity/investment fund shares are the largest category of household financial wealth (39%), followed by life insurance and pensions (38%) and bank deposits (~20%). Mortgage loans account for about 65% of household financial liabilities.

LO2 · Three Classes of Financial Assets

- **Fixed-Income (Debt) Securities** — promise a fixed or formula-determined income stream. Range from very short-term money market instruments (like T-bills, with minimal price risk) to longer-term instruments (bonds), whose prices fluctuate more with interest rates.
- **Equity (Common Stock)** — an ownership share in a corporation. No fixed payment is promised; holders are residual claimants sharing in the firm's success or failure. Riskier than debt, with higher potential return.
- **Derivative Securities** — options and futures contracts whose payoffs depend on the prices of other assets (stocks, bonds, commodities, indexes). Used to hedge risk or to speculate. Commodity and derivative markets let firms adjust exposure to business risks — e.g., a construction firm can lock in the price of copper using futures.

Key insight: moving from fixed income → equity → derivatives generally means moving toward higher potential return and higher risk.

LO3 · Financial Markets and the Economy

Even though financial assets do not create wealth directly, financial markets perform four crucial roles in a developed economy:

1. The Informational Role

Stock prices reflect investors' collective assessment of a firm's prospects. A more optimistic market pushes the price up, which lowers the firm's cost of raising capital and directs investment toward the applications with the greatest perceived potential. Markets may occasionally misallocate (e.g., the dot-com bubble), but no one knows in advance which ventures will succeed — mistakes are inevitable in any allocation system, market-based or centrally planned.

2. Consumption Timing

Financial assets let individuals shift purchasing power across time: high earners save by buying financial assets; retirees sell them to fund consumption. This separates consumption decisions from the constraints of current income.

3. Allocation of Risk

Financial markets let investors with the greatest risk tolerance bear more risk, while conservative investors bear less. If a firm raises capital by selling both stocks and bonds, risk-tolerant investors buy the stock (bearing most of the business risk for potentially higher reward) while conservative investors buy the bonds (fixed payment, lower risk).

4. Separation of Ownership and Management

Large modern corporations (e.g., General Electric, with ~\$493B in total assets) cannot be owner-operated. Shareholders elect a board of directors, which hires and supervises management. This lets ownership change hands (via share sales) without disrupting operations — but it also creates a gap between the interests of owners and managers.

Agency Problems

Managers, hired as agents of shareholders, may pursue their own interests instead — e.g., empire-building, avoiding risky-but-valuable projects to protect their own jobs, or overconsuming perks like corporate jets. These conflicts are called **agency problems**.

Mechanisms That Mitigate Agency Problems

- **Compensation tied to firm performance** — stock and option-based pay means managers only do well if shareholders do well. (Note: overuse of options creates its own agency problem — managers may be tempted to manipulate information to prop up the stock price temporarily so they can cash out before it corrects.)
- **Boards of directors** — increasingly willing to force out underperforming management teams.

- **Outside monitors** — security analysts and large institutional investors (e.g., the Canadian Coalition for Good Governance, whose members collectively manage \$3 trillion) monitor firms closely.
- **The takeover threat** — proxy contests by unhappy shareholders rarely succeed, but underperforming firms can be acquired by other firms that replace management with their own team.

Corporate Governance and Corporate Ethics

Market signals only allocate capital efficiently if investors act on accurate information — markets must be **transparent**. 2000–2002 saw a wave of governance scandals: **WorldCom** overstated profits by at least \$3.8 billion by misclassifying expenses as investments, triggering (at the time) the largest bankruptcy in U.S. history. **Enron** used off-balance-sheet “special-purpose entities” to hide debt. In Canada, accounting issues at **Nortel Networks** (once the largest stock by market cap on the S&P/TSX) led to its bankruptcy. **Parmalat** (Italy) claimed a \$4.8 billion bank account that turned out not to exist.

Other conflicts of interest: stock analysts issued misleadingly optimistic research in exchange for their firms’ investment banking business; IPO shares were allocated to executives as favours. **Auditors** were compromised too — Enron’s auditor Arthur Andersen earned more from consulting for Enron than from auditing it, giving it an incentive to be lenient.

In response, the U.S. passed the **Sarbanes-Oxley Act** (2002): more independent directors, CFOs must personally vouch for accounting statements, an oversight board for auditors, and a ban on auditors providing other services to the same client. Canadian provincial securities regulators implemented comparable reforms.

LO4 · The Investment Process

A portfolio is a collection of investment assets. Investors make two central decisions when building one:

- **Asset allocation** — the choice among broad asset classes (stocks, bonds, real estate, safe money-market assets, etc.), including how much to place in safe versus risky assets.
- **Security selection** — the choice of which particular securities to hold within each asset class.

Important distinction: saving is not the same as safe investing. Saving means not spending all current income; you may then invest those savings in safe assets, risky assets, or a mix of both.

Top-Down vs. Bottom-Up Portfolio Construction

Top-down: start with asset allocation — e.g., decide what proportion to move from cash into stocks vs. bonds — before selecting individual securities. Since 1956, Canadian common stocks have averaged better than 9% per year versus about 5.5% for Canadian T-bills, but with annual returns ranging from –33% to +45% (vs. T-bills, which are essentially risk-free). This decision has major consequences for portfolio risk and return.

Bottom-up: build the portfolio from whichever individual securities look attractively priced, with less regard for the resulting asset mix. This can create unintended concentration in one industry, region, or risk factor — but keeps the focus on the most attractively priced opportunities.

LO5 · Markets Are Competitive

Thousands of well-resourced analysts constantly scour markets for the best buys. This competition implies we should rarely find “free lunches” — securities so underpriced they are obvious bargains. Two key implications follow:

The Risk–Return Trade-Off

Actual returns almost always deviate from what was expected. In 2008 (the worst calendar year for Canadian stocks since 1957), the S&P/TSX Composite Total Return Index fell 33%; in 1979 (the best year), it gained 45%. If a high-return, low-risk asset appeared, investors would rush to buy it, bidding its price up until its expected return fell back to a level commensurate with its risk. The result: **higher-risk assets are priced to offer higher expected returns than lower-risk assets**. The precise quantitative trade-off between risk and return is the subject of modern portfolio theory (Part II of the text), pioneered by Nobel laureates Harry Markowitz and William Sharpe.

Efficient Markets

A second implication: security prices should rapidly reflect all available information (the **efficient market hypothesis**, covered fully in Chapter 11). If true, there are neither underpriced nor overpriced securities on average. This motivates the choice between **passive management** (hold a diversified portfolio without trying to beat the market) and **active management** (try to identify

mispriced securities or time asset classes). If markets were perfectly efficient, active analysis would be pointless — but without anyone doing that analysis, prices would drift away from fair value, creating openings for diligent investors. In practice, markets are probably only *nearly* efficient.

LO6 · The Players

From a bird's-eye view, there are three major participants in financial markets:

- **Firms** — net demanders of capital. They raise funds now to invest in plant and equipment.
- **Households** — typically net suppliers of capital, purchasing the securities firms issue.
- **Governments** — can be borrowers or lenders depending on the gap between tax revenue and spending. The Canadian federal government ran deficits from the 1960s to the mid-1990s, then surpluses from 1998–2008, and has run deficits again since.

Financial Intermediaries

Corporations and governments do not sell most of their securities directly to individual households. About half of all stock is held by large institutions (pension funds, mutual funds, insurance companies, banks) that stand between the issuer and the ultimate investor — called **financial intermediaries**. A bank, for example, raises funds by taking deposits and lends that money out; the spread between deposit and lending rates is the bank's profit. In Canada, real assets account for only about 1% of chartered banks' total assets (versus over 51% for non-financial companies) — intermediaries' assets and liabilities are overwhelmingly financial.

Intermediaries add value by pooling many small investors' funds (enabling large loans and diversification) and by building risk-assessment expertise through volume. **Mutual funds** pool retail money for diversified, professionally managed portfolios. **Hedge funds** do something similar but are restricted to institutional/wealthy investors, pursue more complex and higher-risk strategies, and typically charge a share of trading profits rather than a flat asset-based fee.

Investment Bankers

Firms that need to raise capital use **investment bankers** (underwriters) rather than maintaining an in-house issuance function, since public offerings happen infrequently. Investment bankers advise on pricing and rates, then market the security in the **primary market**. Afterward, investors trade the security among themselves in the **secondary market**.

In the U.S., the Glass-Steagall Act (until 1999) legally separated commercial and investment banking; Canada removed the equivalent separation much earlier, in 1987, so most Canadian banks were already “universal banks.” Most of Europe never separated the two. By September 2008, all remaining major U.S. stand-alone investment banks (Bear Stearns, Merrill Lynch, Lehman Brothers, and finally Goldman Sachs/Morgan Stanley) were absorbed into commercial banks, went bankrupt, or converted into bank holding companies — ending the era of the independent investment bank. The debate resurfaced later via the Dodd-Frank Act's Volcker Rule, which restricts banks from proprietary trading and limits their investments in hedge funds and private equity.

Venture Capital and Private Equity

Young, private firms that have not yet issued public securities rely on bank loans and **venture capital (VC)** — equity investment from dedicated VC funds, wealthy “angel investors,” or institutions. Most VC funds are structured as limited partnerships: a management company invests its own money plus capital from limited partners (e.g., pension funds), sits on the start-up's

board, helps recruit senior managers, and eventually liquidates the fund (e.g., after ~10 years) to distribute proceeds. Related **private equity** investors take similar active, hands-on roles but focus on distressed or under-managed firms that can be bought, improved, and resold. Collectively, these investments in companies that do not trade on public exchanges are called private equity investments.

LO7 · The Financial Crisis of 2008

The 2008 crisis illustrates almost every concept in this chapter — agency problems, securitisation, and systemic risk — and shows the intimate link between the financial system and the real economy.

Antecedents of the Crisis

After the 2000–2002 dot-com bust, the Fed cut interest rates aggressively; T-bill and LIBOR rates fell sharply, and the recession was short and mild. By 2007 the TED spread (LIBOR minus T-bill rate, a gauge of banking-sector credit risk) was only about 0.25%, suggesting minimal fear of bank defaults — a period some called the “Great Moderation.” Cheap credit fed a historic U.S. housing boom: average U.S. house prices more than doubled over the decade beginning in 1997.

Changes in Housing Finance

Historically, local lenders held the mortgages they originated. This changed once Fannie Mae and Freddie Mac began buying mortgages and bundling them into tradeable mortgage-backed securities (MBS) — the “originate-to-distribute” model. Conforming loans (meeting strict underwriting criteria, e.g., loan-to-value $\leq 80\%$) were pooled mainly through these agencies. But the model also opened a niche for private firms to securitize **non-conforming subprime loans**, where the investor — not the agency — bore the default risk. Because originators could sell loans immediately, they had little incentive to screen borrowers carefully; low- and no-documentation (“liar”) loans proliferated, and loan-to-value ratios rose to the point where many subprime borrowers financed the entire purchase price. Adjustable-rate mortgages with low “teaser” rates that later reset sharply upward added further risk. Once housing prices peaked in 2006, refinancing options dried up and default rates surged in 2007.

Mortgage Derivatives: CDOs

Investment banks used **collateralized debt obligations (CDOs)** to carve AAA-rated securities out of pools of risky loans by dividing the pool into senior and junior **tranches**. Example: if 70% of a pool is senior and 30% junior, senior investors are repaid in full as long as the default rate stays below 30% — which seemed very safe, so rating agencies (Moody’s, S&P, Fitch) awarded senior tranches AAA ratings. These ratings proved far too optimistic: they were based on historical data from an unrepresentative housing boom, assumed diversification across regions would protect the pool (which failed once prices fell nationwide simultaneously), and were paid for by the very issuers seeking favourable treatment — an agency problem in the rating process itself.

Credit Default Swaps

In parallel, the market in **credit default swaps (CDS)** — essentially insurance contracts against borrower default — exploded. Some swap issuers took on far more credit-risk exposure than they had capital to support. The insurer **AIG** alone sold more than **\$400 billion** of CDS contracts on subprime mortgages.

Systemic Risk and “The Shoe Drops”

Many large financial institutions were funding long-term illiquid assets with short-term borrowing, at high leverage with thin capital buffers — a fragile model requiring constant refinancing. Widespread reliance on hard-to-value, illiquid CDOs, and the shift from transparent exchange trading (with daily-margined positions) to opaque over-the-counter CDS contracts, created **systemic risk**: the danger that trouble in one market spills over and disrupts others.

The crisis peaked in September 2008: Fannie Mae and Freddie Mac were put into conservatorship (Sept. 7); Merrill Lynch was sold to Bank of America (Sept. 14); Lehman Brothers filed the largest bankruptcy in U.S. history after being denied a rescue (Sept. 15); the government lent AIG \$85 billion (Sept. 17) to prevent its collapse from destabilizing the banks holding its CDS guarantees; and the Treasury proposed a \$700 billion toxic-asset purchase program (Sept. 18). The **Reserve Primary Money Market Fund** “broke the buck” (its NAV fell below \$1/share) after losses on Lehman commercial paper, triggering a run on money market funds and freezing short-term credit markets — which then choked off financing for businesses well beyond Wall Street, driving unemployment up sharply.

The Dodd-Frank Reform Act (2010)

- **Capital & liquidity rules** — stricter requirements, especially for banks large enough that their failure threatens the system.
- **Derivatives transparency** — real-time reporting; many CDS contracts standardized onto exchanges with daily margining.
- **The Volcker Rule** — prohibits banks from proprietary trading and limits their hedge-fund/private-equity investments.
- **Incentive reforms** — “claw-back” provisions let firms recover executive pay based on inaccurate financials.
- **Ratings-agency oversight** — a new SEC Office of Credit Ratings to address the issuer-pays conflict.

The Canadian Experience

Canada was not immune — the S&P/TSX Total Return Index fell almost 50% from its peak and real GDP fell more than 4% — but the impact was ultimately milder than elsewhere, with **no failures of Canadian banks**. In 2008 the World Economic Forum ranked Canadian banks the **soundest among 134 countries**. Contributing factors: Canadian banks were better capitalized, followed more conservative lending practices, kept a larger share of mortgages on their own balance sheets (funded by stable deposits rather than securitization), and were already “universal banks” combining commercial and investment banking. On the housing side, subprime lending was limited, regulations required mortgage-default insurance for homeowners with less than 20% equity, and (unlike in the U.S.) mortgage interest is not tax-deductible in Canada, removing an incentive to take on more mortgage debt. As a result, Canadian housing prices fell only 8.5% from peak and had fully recovered by the end of 2009.

- Real assets create wealth; financial assets distribute claims to that wealth and net to zero across the economy.
- Financial markets improve resource allocation, enable consumption timing, allocate risk to those most willing to bear it, and allow separation of ownership from management.
- Agency problems are pervasive; compensation design, board oversight, monitoring, and regulation mitigate — but never fully eliminate — them.
- Higher expected return always requires accepting higher risk; competitive markets rarely offer "free lunches."
- The financial system has three core participants (firms, households, governments) linked by intermediaries, investment bankers, and VC/private equity providers.
- The 2008 crisis showed how misaligned incentives at every step — originators, raters, insurers, regulators — can compound into systemic risk with real economic consequences.
- Canada's more conservative banking and mortgage regulations produced a meaningfully milder crisis experience than the U.S.

Chapter 1 is primarily conceptual. These are the framework relationships that underpin later quantitative chapters.

F1 | Net National Wealth

National Wealth = Sum of All Real Assets (financial assets net to zero)

Every financial asset is simultaneously someone's asset and someone else's liability across the whole economy. Only real assets constitute national wealth.

F2 | Risk–Return Principle

Higher Expected Return $\Leftrightarrow$ Higher Risk (no free lunch)

Competition drives prices until each asset offers a return commensurate with its risk. A high-return, low-risk opportunity attracts buying that raises price and lowers expected return until the trade-off is restored.

F3 | Equivalent Taxable Yield (preview — formalised in Ch. 2)

$r_{\text{taxable}} = r_{\text{muni}} / (1 - t)$

r_{muni} = tax-exempt municipal yield; t = investor's marginal tax rate. Lets an investor compare a tax-exempt bond to a taxable one on an apples-to-apples basis. (Formally introduced and applied in Chapter 2.)

F4 | CDO Tranche Loss Threshold

Senior tranche is repaid in full as long as: Pool Default Rate < Junior Tranche %

Example: a 70% senior / 30% junior split means the senior tranche is fully protected unless more than 30% of the underlying loans default — which seemed extremely unlikely under historical (pre-2008) assumptions.

EXAMPLE 1.1 | Activist Investors and Corporate Control

Context

Illustrates how large shareholders act as a market-based mechanism to reduce agency problems: activists buy large stakes and use that ownership to pressure management. Ties directly to LO3 (agency problems and corporate governance).

Activist Investors and Their Initiatives (textbook list)

Investor / Fund	Initiative
Carl Icahn	One of the earliest and most combative activist investors. Challenged Apple to increase cash distributions to investors.
William Ackman, Pershing Square	Took large positions in Canadian Pacific Railway, JCPenney, Valeant Pharmaceuticals, and Kraft Foods to influence management practice.
Nelson Peltz, Trian	Sought board seats at DuPont; pushed for it to split into more focused corporations.
Dan Loeb, Third Point	Tried to get Sony to spin off its entertainment units.
Jeff Smith, Starboard Value	Pushed for Staples and Office Depot to merge; the firms attempted it, but the merger was blocked on antitrust grounds.

What This Example Teaches

Large investors have both the incentive (a significant stake makes monitoring worthwhile) and the means (voting power, board influence) to hold management accountable — a market-based solution to the agency problem that dispersed small shareholders cannot achieve individually.

- **Agency problem:** managers may not act in shareholders' best interest.
- **Monitoring as governance:** concentrated ownership creates the incentive to monitor that dispersed ownership lacks.
- **Secondary market liquidity:** activists can only acquire large stakes because secondary markets allow free share trading.
- **External constraints:** even privately agreed strategic changes can be blocked by regulators (Staples/Office Depot).

PRACTICE PROBLEM | MapleGrow Ltd. — Putting It All Together

Scenario

MapleGrow Ltd. is a Canadian agricultural technology company raising \$50 million, choosing between:

- **Option A — Equity:** 1,000,000 new common shares at \$50 each.
- **Option B — Debt:** 5-year bonds, \$50 million face value, 5% fixed coupon.

MapleGrow has 4,000,000 shares outstanding. The CEO owns 200,000 shares and is paid \$500,000 base salary plus options exercisable at \$50/share. Activist fund **Grizzly Capital** recently acquired 15% of outstanding shares and is publicly lobbying the board to replace the CEO.

Part A — Real vs. Financial Assets

Classify each item as a real asset, a financial asset, or neither, with brief reasoning:

- A fleet of agricultural drones purchased last year.
- The 1,000,000 new common shares MapleGrow plans to issue.
- A patent on MapleGrow's soil-sensing algorithm.
- A futures contract to sell wheat at a fixed price in 6 months.
- Cash in a chequing account at TD Bank.

Part B — Primary vs. Secondary Market

MapleGrow hires an investment bank to manage the Option A offering.

- Is this a primary or secondary market transaction? Explain.
- Who receives the proceeds?
- Where will the shares trade afterward, and what do we call that market?

Part C — Fixed Income vs. Equity Risk

- Under what circumstances would the bond (Option B) outperform for an investor?
- Under what circumstances would the equity (Option A) outperform?
- In bankruptcy, who is paid first — bondholders or shareholders? Why?

Part D — Agency Problems & Governance

Identify at least two distinct agency problems present in this scenario.

For each, describe the conflict of interest and suggest a governance mechanism (from LO3) that could mitigate it.

Part E — The Activist Investor

- (i) What role is Grizzly Capital playing here, in the LO3/LO6 sense?
- (ii) Is acquiring the 15% stake a primary or secondary market activity?
- (iii) Does Grizzly's behaviour reduce or worsen the agency problem? Explain, referencing Example 1.1.

Part F — Risk–Return Trade-Off

MapleGrow shares are priced to offer an expected 12% return; the bonds offer 5%.

An investor claims the shares are "mispriced — a free 7% extra return."

Evaluate this claim using the LO5 risk–return trade-off. What would rational investors do if the gap were truly free?

Hints & Guidance

Part A tests LO1–LO2. Parts B–C test LO3–LO6. Part D tests LO3. Parts E–F synthesise LO3, LO5, and LO6. For Part F, ask: if the 7% premium were truly free, what would thousands of rational investors do to the share price?